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Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 12	
TITLE : User-Defined Packages	

Problem Statement

Write a Java program to create and use user-defined packages.

Learning Objectives

To create and use user-defined packages for organizing Java programs.

Theory

Packages are used to organize related classes and interfaces into namespaces. User-defined packages improve code organization, reduce naming conflicts, and enhance reusability. They also support access protection and modular application development. Java packages simplify project management by grouping logically related components together. Large-scale software systems extensively use packages.

Programme

1. Create a Java program by developing user-defined packages for a College Management System. Create separate packages for student and faculty details, and display their information using classes from these packages.

```
package student;

public class Student {

    public void displayStudent() {
        System.out.println("Student Name: Anurag ");
        System.out.println("Roll Number: 101");
        System.out.println("Course: B.Tech");
    }
}
```

```
package faculty;

public class Faculty {

    public void displayFaculty() {
        System.out.println("Faculty Name: Dr. Amit");
        System.out.println("Faculty ID: F101");
        System.out.println("Department: Computer Science");
    }
}
```

```
import student.Student;
import faculty.Faculty;

public class CollegeManagement {

    public static void main(String[] args) {

        Student s = new Student();
        Faculty f = new Faculty();

        System.out.println("Student Details:");
        s.displayStudent();

        System.out.println();

        System.out.println("Faculty Details:");
        f.displayFaculty();
    }
}
```

2. Create a Java program by developing a **user-defined package** named library that contains a class to store and display book details such as book ID, title, author, and price. Import the package into the main program and display the book information.

```
package library;

public class Book {

    int bookId;
    String title;
    String author;
    double price;

    public Book(int bookId, String title, String author, double price) {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
        this.price = price;
    }

    public void displayBook() {

        System.out.println("Book ID: " + bookId);
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Price: Rs. " + price);
    }
}
```

```
import library.Book;

public class LibraryManagement {

    public static void main(String[] args) {

        Book b = new Book(
            101,
```

```

        "Java Programming",
        "Herbert Schildt",
        650
    );

    System.out.println("Book Details:");
    b.displayBook();
}
}

```

Output

```

● rakeshkumarchaudhary@RAKESHs-MacBook-Air assignment13 % javac Employee.java
● rakeshkumarchaudhary@RAKESHs-MacBook-Air assignment13 % java Employee
Employee details written successfully.

Employee Details:
Employee ID: 101
Employee Name: Anurag
Department: Computer Science
Salary: 40000

○ rakeshkumarchaudhary@RAKESHs-MacBook-Air assignment13 % █

```

```

cd "/Users/rakeshkumarchaudhary/LibraryManagement"
cd "/Users/rakeshkumarchaudhary/LibraryManagement/" && javac LibraryManagement.java &&
● rakeshkumarchaudhary@RAKESHs-MacBook-Air LibraryManagement % cd "/Users/rakeshkumarcha
● rakeshkumarchaudhary@RAKESHs-MacBook-Air LibraryManagement % cd "/Users/rakeshkumarcha
&& java LibraryManagement
Book Details:
Book ID: 101
Title: Java Programming
Author: Herbert Schildt
Price: Rs. 650.0
○ rakeshkumarchaudhary@RAKESHs-MacBook-Air LibraryManagement %

```

Conclusion

In this practical, I learned how to create and use **user-defined packages in Java**. I created separate packages for student, faculty, and library details and imported their classes into the main program. This helped me understand how packages organize related classes and make Java programs easier to manage, reuse, and maintain.